

Design Evaluation 2: XR Study Whiteboard Iteration 2

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Objective and Validation Metrics

Iteration 2 was tested to see whether a student could complete one clear VR classroom activity. The student needed to move to a desk, view the board and paper together, take notes, use the paper tools, return to the whiteboard, and replay a recorded board action. These tasks addressed the Round 1 problems with the seated view, paper controls, controller movement, and unclear navigation. The evaluation measured usability and comfort. It did not test learning outcomes.

Table 1
Success Measures Used for Iteration 2

Area	Success measure
Shared seated view	At least four of five can see the board and paper from the desk.
Paper use	At least four of five can select each tool and make the required marks.
Navigation	Controllers stay aligned and median predictability is at least 4.
Recording	At least four of five complete and explain Record, Stop, and Play.
Confidence and safety	Median confidence is at least 4 with no serious comfort, tracking, or data loss issue.

Method

Five university students completed the same moderated task in class with the Unity VR build, a headset, and two physical controllers. They explored the room, copied board content onto paper, used both drawing surfaces, moved between locations, and tested replay. P01 had a voice recording and P02 had a camera recording. P03 to P05 were not recorded, so their evidence comes from observation notes and ratings. The simulator T and Y task was excluded because every participant used physical controllers.

Results

All five participants completed every observed headset task. This included the shared seated view, note copying, paper tools, movement, whiteboard tools, replay, and controller tracking. No session stopped because of discomfort or a technical issue. The full task matrix is in Appendix A.

Analysis and Insights

Table 2
Post-Test Ratings

Measure	P01	P02	P03	P04	P05	Median
Seated view comfort	5	5	5	5	5	5
Paper drawing reliability	5	5	5	5	5	5
Paper control clarity	5	5	5	5	5	5
Movement predictability	5	5	5	5	5	5
Recording control clarity	5	5	5	5	5	5
Overall confidence	5	5	5	5	5	5

Note. Ratings used a scale from 1, very poor, to 5, excellent.

The strongest result was that both study surfaces worked as one activity. All five participants viewed the teaching board and wrote on their paper without leaving the desk. P01 called the task “pretty easy,” and P02 said the shared view was comfortable. Everyone also wrote, erased, and cleared the paper. Fixing the paper and using the right controller for writing removed the earlier confusion between grabbing and writing.

The guide supported first use. P02 noticed it at once and opened it again. P01 needed a short time to learn the interaction, and P04’s notes suggested a brief introduction. Replay was clear after use. All five participants completed Record, Stop, and Play. P01 expected the marks and timing to repeat, while P02 said replay matched her expectation. This supports action replay without audio.

Evaluation of the Testing Aims

The shared view, navigation, tracking, recording, confidence, and safety aims were validated by five task passes and median ratings of 5. Paper use was strongly supported, but the planned 15-second discovery time and exact 90% stroke target cannot be claimed because timing, prompts, and stroke defects were not recorded. Simulator controls were not evaluated. The evidence therefore validates the tested headset workflow only.

Concept Iteration

The core design should remain because no repeated problem blocked the task. The left controller handles movement, X moves to the whiteboard, and Y moves through the tables. The right trigger selects tools and writes. The fixed paper, seated viewpoints, tracking, and replay should remain. The guide should open at the start, close with A, and stay available later to help first-time users.

Updated Concept Description

XR Study Whiteboard is a virtual classroom for remote study. A learner follows the guide, moves between the whiteboard and desks, and uses the right controller to write. At a desk, the teaching board and fixed paper remain visible together. The learner can take notes, use drawing tools, and replay whiteboard actions. This gives remote learners one clear classroom workflow.

Reflection and Next Steps

The same task order made sessions easy to compare, and two recordings showed how participants understood the controls. Three sessions only have notes, the moderator gave some prompts, and every rating reached the highest score. Discovery time, prompt level, and stroke defects were not recorded. The next test should use five new participants with only the in-headset guide. It should record the missing measures and use a longer lesson to test note taking and saving. Simulator controls should be tested separately.

Final Evaluation

Iteration 2 is validated for the tested headset workflow. Five of five participants met every applicable core target with no serious issue. The interaction design should be kept while the next study tests independent first use.

Appendix A

Evidence Register and Participant Results

All five participants used the Iteration 2 VR build with a headset and two physical controllers. Only P01 and P02 were recorded. P03 to P05 are notes-only records and do not contain direct quotations.

Table 3

Evidence Available for Each Participant

ID	Participant	Evidence	Outcome
P01	Kai	Voice recording and transcript	Completed
P02	Syifa	Camera recording and transcript	Completed
P03	Amy	In-class notes only	Completed
P04	Serafina	In-class notes only	Completed
P05	Safira	In-class notes only	Completed

Table 4

Participant Task Matrix

Task	P01	P02	P03	P04	P05
Explore	Pass	Pass	Pass	Pass	Pass
Shared seated view	Pass	Pass	Pass	Pass	Pass
Follow and take notes	Pass	Pass	Pass	Pass	Pass
Paper tools and strokes	Pass	Pass	Pass	Pass	Pass
Move and return	Pass	Pass	Pass	Pass	Pass
Whiteboard workflow	Pass	Pass	Pass	Pass	Pass
Record and replay	Pass	Pass	Pass	Pass	Pass
Physical controller tracking	Pass	Pass	Pass	Pass	Pass
Simulator T and Y task	N/A	N/A	N/A	N/A	N/A

Recording Links

P01 voice recording: [Google Drive evidence](#)

P02 camera recording: [Google Drive evidence](#)

Appendix B

Notes-Only Session Summaries

P03: Amy

Amy completed all headset tasks. The notes record stable navigation, correct use of Pencil, Eraser, and Clear Paper, successful Record, Stop, and Play, and correct controller tracking. Every post-test measure was 5 out of 5. Overall satisfaction was 10 out of 10. Optional extra classroom features were suggested, but no core usability change was requested.

P04: Serafina

Serafina completed all headset tasks. The notes record a comfortable shared view, stable controller tracking, successful drawing, and successful replay. Every post-test measure was 5 out of 5. Overall satisfaction was 10 out of 10. A short introduction for first-time VR users was the only suggested change.

P05: Safira

Safira completed all headset tasks. The notes record a comfortable shared view, reliable paper and whiteboard use, stable movement, and successful replay. Every post-test measure was 5 out of 5. Overall satisfaction was 10 out of 10. Saved notes and more classroom environments were suggested as optional future features.

Appendix C

Recorded Interview Transcripts

These transcripts reproduce the two recorded sessions. A moderator note is retained where P01 referred to a mouse even though the session used a headset and physical controllers.

P01: Kai, Voice-Recorded Transcript

Nayaka: Good afternoon. My name is Nayaka, and I am going to guide you through testing my XR Study Whiteboard. Before we start, may I know your name?

Kai: Yeah, my name is Kai.

Nayaka: Nice to meet you.

Kai: Nice to meet you too.

Nayaka: Before we start, you are using the Unity VR build with the headset and two physical controllers. Can you see both controllers in the environment?

Kai: Yeah, I can see both of them and they are following my hands.

Nayaka: First, can you tell me what you see in the XR environment?

Kai: It looks like I am in a classroom right now. I can see a whiteboard. It says XR Study Whiteboard and Study Session. I can also see black, red, marker, eraser, Ready to Record, and Play.

Nayaka: Can you see all the buttons?

Kai: Yeah. Whiteboard, Stand, Jump, Center Table.

Nayaka: Can you try pressing one of them?

Kai: Yeah.

Nayaka: What do you think these buttons do? What were you expecting?

Kai: Initially, I did not know what to do. But now that I have pressed it, I can see that it teleports me to one of the tables depending on which one I press. If I press Stand, it puts me back.

Nayaka: When you move, do both VR controllers stay in the correct position with your hands?

Kai: Yeah, both of them stay with me.

Nayaka: Okay. Can you move to the whiteboard?

Kai: Yeah.

Nayaka: After teleporting to the whiteboard, what do you think you can do here?

Kai: Looking at the marker and everything, I expect that I could act like a teacher and write something on the whiteboard.

Nayaka: Okay. Can you try it?

Kai: Hello.

Nayaka: What do you think the eraser does?

Kai: I imagine it erases whatever I just wrote.

Nayaka: Nice. What other controls can you see?

Kai: Clear Board.

Nayaka: What do you expect that to do?

Kai: Clear my whole board.

Nayaka: Yes, that is right. What about changing colours?

Kai: Yeah, I can change the colour.

Nayaka: Do you see any other buttons?

Kai: There is a Record button.

Nayaka: What do you think that does?

Kai: It says Ready to Record, and that it records board actions and timings. I am assuming it records what I write. Then when I press Play, it will repeat what I wrote.

Nayaka: Can you try it?

Kai: For example, if I write “Hello”, then press Stop and Play, I would imagine it clears the board first and then writes “Hello” again.

Nayaka: Yes, that is right.

Kai: Oh, okay.

Nayaka: Great. We have finished testing the whiteboard. Can you now try the features on the desk?

Kai: Yep. Now I am at one of the desks. I am pressing the left click button on the paper to write “Hello”. I am going to press Eraser, and I expect it to erase what I wrote.

Nayaka: Yes.

Kai: Now I am going to write a lot on the paper. Then I will press Clear Paper. I expect it to clear the paper.

Nayaka: Okay. While you are at the desk, can you see the paper and the teaching board comfortably from the seated position?

Kai: Yeah. I can look between the paper and the board without needing to move away from the desk.

Nayaka: Can you copy the date, horizontal line, and circle from the teaching board onto the paper?

Kai: Yeah.

Nayaka: How did that feel?

Kai: Pretty easy. I can look at the board and then write it on the paper.

Nayaka: Okay. What do you think the overall purpose of this prototype is, and who do you think would use it?

Kai: I think it is like a digital classroom. It could be for students or teachers. Maybe it would be useful if something similar to COVID happened again and people had to learn from home.

Nayaka: Yes. One of the ideas behind this is that during COVID, students could sometimes find it difficult to stay focused when classes were only through something like Zoom. With this, students could feel more like they are actually inside a classroom.

Kai: Yeah.

Nayaka: From one to five, how difficult do you think the prototype is to use?

Kai: It is not very difficult because I am using a mouse. If I was using a VR controller, I do not think it would be much harder either. It is pretty easy overall.

Moderator note: The session itself was completed using the VR headset and physical controllers. The wording above is retained from the original recording transcript.

Nayaka: Thinking specifically about the VR controllers you are using now, did they feel easy to control?

Kai: Yeah. They are easy to use and both controllers respond properly.

Nayaka: Would you use something like this if you were in a situation where you needed to attend classes remotely?

Kai: Yeah. The idea is cool, especially because you are inside a classroom instead of being alone at home. I think I would definitely use it.

Nayaka: Thank you. Do you have any feedback or anything you think I should improve?

Kai: It seems pretty good. I think everything is pretty clear. I am just trying to think of something. The writing and deleting controls are clear. I do not really have anything else to add. It is pretty good.

Nayaka: Okay. Thank you so much for your time.

Kai: No worries.

P02: Syifa, Camera-Recorded Transcript

Nayaka: Okay, I am going to do VR testing. May I know your name?

Syifa: Syifa.

Nayaka: Can I record this interaction?

Syifa: Yeah.

Nayaka: Okay. First, can you grab the two controllers? What do you see on the screen?

Syifa: There is a pop-up guide.

Nayaka: Before we continue, can you see both physical controllers inside the VR environment?

Syifa: Yeah.

Nayaka: Are both controllers following your hands correctly?

Syifa: Yeah.

Nayaka: What does it say?

Syifa: Controls.

Nayaka: Okay. Press A. The A button is on your right controller. What can you understand from the guide?

Syifa: There are instructions for the controls.

Nayaka: So you understand the guide?

Syifa: Yeah.

Nayaka: Can you move from one table to another table?

Syifa: Like this?

Nayaka: Yes. Maybe close the guide first and then try moving to another table.

Syifa: Okay.

Nayaka: Did both physical controllers stay aligned with your hands when you moved?

Syifa: Yeah.

Nayaka: Can you look around? What do you see on the table?

Syifa: Paper.

Nayaka: What else?

Syifa: The instructions.

Nayaka: Can you also see the teaching board from where you are sitting?

Syifa: Yeah.

Nayaka: Can you look between the teaching board and the paper comfortably?

Syifa: Yeah.

Nayaka: Can you copy the date, horizontal line, and circle from the board onto your paper?

Syifa: Okay.

Nayaka: Was that comfortable?

Syifa: Yeah.

Nayaka: Can you try writing something on the paper? You can look at the guide again if you need help.

Syifa: Okay.

Nayaka: Now try changing the tool to the eraser.

Syifa: Okay.

Nayaka: Can you erase what you wrote?

Syifa: Yeah.

Nayaka: What other button can you see?

Syifa: Clear.

Nayaka: Can you try clearing the paper?

Syifa: Okay.

Nayaka: Now can you move to the whiteboard?

Syifa: Okay.

Nayaka: Nice. I can see you are looking at the guide again. You can close it when you are ready. What do you see on the whiteboard?

Syifa: The tools and colours.

Nayaka: You can move around using both controllers. Try moving forward. Now can you write something on the whiteboard using the red marker? You can check the guide again if you are confused.

Syifa: Okay.

Nayaka: Nice. What else can you see on the whiteboard interface?

Syifa: The colours.

Nayaka: What else?

Syifa: The tools. Eraser.

Nayaka: Anything else?

Syifa: Record.

Nayaka: Can you try the Record button? Before that, can you clear the whiteboard?

Syifa: Okay.

Nayaka: Now press Record and write something on the whiteboard. Maybe use the blue marker. Select Marker first, then select Blue.

Syifa: Okay.

Nayaka: Nice. Now press Stop and then try the Play button.

Syifa: Okay.

Nayaka: Let us try it again. Draw something, but make sure you press Record first. After you finish, press Stop, and then press Play.

Syifa: Okay.

Nayaka: What do you understand from this activity?

Syifa: We can write something on the whiteboard, draw something on the paper, and record it as well.

Nayaka: Did the replay match what you expected?

Syifa: Yeah.

Nayaka: And while moving and drawing, did the left and right VR controllers stay responsive?

Syifa: Yeah, both of them worked properly.

Nayaka: Yes. The purpose is to create a virtual classroom so that students who are studying remotely, such as through Zoom, can feel more like they are inside a real classroom and hopefully pay more attention to the teacher. Would you use something like this if you were studying remotely, like during COVID?

Syifa: Yeah.

Nayaka: How would you rate the interaction activities?

Syifa: Ten.

Nayaka: Ten out of ten?

Syifa: Yeah.

Nayaka: Do you have any feedback or anything you think should be improved?

Syifa: No, it is already good.

Nayaka: Okay. Thank you for your time.

Appendix D

Evidence Limits and Success Measure Audit

Table 5
Audit of the Planned Measures

Measure	Evidence	Status
Shared seated view	Five passes and five ratings of 5	Met
Paper tool discovery	Five passes. Exact time and prompt level were not recorded	Timing not verified
Paper drawing	Five passes and five ratings of 5. Defect counts were not recorded	Exact percentage not verified
Lesson-to-notes task	Five passes and five comfort ratings of 5	Met
Navigation stability	Five passes and five predictability ratings of 5	Met
Whiteboard recording	Five passes. P01 and P02 have recorded evidence	Met
Simulator controls	All sessions used physical controllers	Not evaluated
Overall confidence	Median 5. No serious issue was recorded	Met

Limits of the Evidence

Only two sessions have audio or video recordings. Three sessions rely on observation notes.

Some moderator prompts may have made the tasks easier.

Every rating reached the highest score, so the scale did not capture smaller differences.

The records do not include exact discovery times, prompt levels, or stroke defect counts.

The small in-class sample supports a usability decision for this build. It does not prove an improvement in learning outcomes.

Traceability

The combined transcript and result record is stored in Design Evaluation/Interview Transcripts/Iteration2_XRStudyWhiteboard_Transcript_and_Results.md. The supplied assessment brief was used only as the task description and marking rubric. It was not treated as participant evidence.